

Exact zero quantum and private capacity of rational channels

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Abstract

A computable complementary-channel completion is given for the uniformly gapped inverse-Jordan kernels produced by the supplied companion channel-order reduction. The completion is a genuine rational Stinespring construction. Its soundness holds for coherent inputs, arbitrary Hermitian tangents, and every finite quantum reference, with no state-conditioning or reference-size bound. An exact purification duality controls all leaf scores, and sector-copy padding absorbs the remaining error. A zero-rate coding argument compiles every finite-dimensional half-erasure helper into finitely many uses of the fixed qubit half-erasure channel. Together with the companion source theorem, the unrestricted complete information-order equivalence, and established capacity bridges, the assembled proof gives the classification: exact zero quantum capacity and exact zero private capacity are co-recursively enumerable complete for finite rational channels, while their positive-capacity complements are recursively enumerable complete. The proof relies on the supplied companion research theorem and has not undergone external review. It is an exact zero-versus-positive result; no capacity approximation, fixed-gap promise, efficient reduction, or computable positive-capacity witness bound is asserted.

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1 Statement and proof dependencies

All Hilbert spaces in individual tests are finite dimensional, but the dimensions of auxiliary references and codes are unrestricted. A rational complex matrix has real and imaginary entries in $\mathbb{Q}$. The input description of a channel is its finite rational complex Choi matrix, with positivity and trace preservation imposed exactly. The construction below in fact supplies rational complex Kraus matrices for its principal channel.

For a channel $\Phi : A \rightarrow B$ and a fixed genuine complement $\Phi^c : A \rightarrow E$, define

$$\begin{aligned}
Q^{(1)}(\Phi) &= \max_{\rho} \{S(\Phi(\rho)) - S(\Phi^c(\rho))\}, \\
P^{(1)}(\Phi) &= \max_{\{p_x, \rho_x\}} \{I(X : B) - I(X : E)\}, \\
Q(\Phi) &= \sup_{n \geq 1} \frac{1}{n} Q^{(1)}(\Phi^{\otimes n}), \quad P(\Phi) = \sup_{n \geq 1} \frac{1}{n} P^{(1)}(\Phi^{\otimes n}).
\end{aligned}$$

Entropies in the capacity formulas use base two. The private ensemble is finite and may contain mixed states entangled across the n uses. These quantities are complement-independent, and $0 \leq Q(\Phi) \leq P(\Phi)$. Indeed, the spectral pure-state ensemble of any ρ has equal conditional receiver and environment entropies, so its private information equals $I_c(\Phi, \rho)$; apply the regularized formulas.

Theorem 1.1 (Exact capacity classification). *Assume the gapped source theorem and unrestricted complete-order equivalence stated in Section 2, as proved in the explicitly supplied companion research manuscript. The following sets of finite rational channels are coRE-complete under computable many-one reductions:*

$$\{\Phi : Q(\Phi) = 0\}, \quad \{\Phi : P(\Phi) = 0\}.$$

Their complements $\{\Phi : Q(\Phi) > 0\}$ and $\{\Phi : P(\Phi) > 0\}$ are RE-complete. More precisely, one effectively constructs from a Turing machine M and its input x a rational channel

$$T_{M,x} = C_{M,x} \otimes \mathcal{E}_{2,1/2} \tag{1}$$

such that

$$\begin{aligned}
M(x) \text{ does not halt} &\implies Q(T_{M,x}) = P(T_{M,x}) = 0, \\
M(x) \text{ halts} &\implies Q(T_{M,x}) > 0 \text{ and } P(T_{M,x}) > 0.
\end{aligned}$$

The helper $\mathcal{E}_{2,1/2}$ in every output instance is the same qubit half-erasure channel.

The companion manuscript *Undecidability of complete quantum channel information order*, dated 10 October 2026, is supplied together with this manuscript [1]. It is a research theorem

with its own proof, not an external publication or a peer-review certificate. Its game-to-kernel construction is the source premise used here; its later additive-approximation result is not used. The public hardness premise behind that source is the finite-dimensional nonlocal-game consequence of $\text{MIP}^* = \text{RE}$ [2]. The present contribution is the physical complementary completion and the fixed-helper compilation. The final capacity implications are assembled in Section 8, and recursive enumerability is proved in Section 11.

2 Information metrics and the supplied source

2.1 Metric conventions and complete quantifiers

For $W > 0$ and a Hermitian matrix Y , set

$$\Omega_W(Y) = \frac{WY + YW}{2}, \quad g_W(Y) = \text{Tr}[Y\Omega_W^{-1}(Y)]. \quad (2)$$

The Hilbert–Schmidt version uses $\text{Tr}[Y^*\Omega_W^{-1}(Y)]$ for arbitrary complex Y . The rectangular Sylvester operator is *unhalved*:

$$\mathcal{J}_{U,V}(Z) = UZ + ZV.$$

Thus $\Omega_W = \mathcal{J}_{W,W}/2$. At a semidefinite base, the metric is given by

$$g_W(Y) = \sum_{a,b:w_a+w_b>0} \frac{2|Y_{ab}|^2}{w_a + w_b}, \quad (3)$$

provided the kernel-to-kernel block of Y is zero; otherwise the extended metric is infinite. Support-to-kernel blocks are permitted. For $t > 0$, $g_{tW}(tY) = tg_W(Y)$.

Write $\Phi \succeq_{\text{SLD}}^c \Psi$ if, for every finite reference R , every faithful input state ρ on $A \otimes R$, and every traceless Hermitian X ,

$$g_{(\Phi \otimes \text{id})(\rho)}((\Phi \otimes \text{id})(X)) \geq g_{(\Psi \otimes \text{id})(\rho)}((\Psi \otimes \text{id})(X)).$$

For faithful input, an output is faithful on the fixed support of the channel’s output on I . There is no uniform reference cutoff. Allowing all Hermitian tangents gives the same comparison: subtract $(\text{Tr } X)\rho$, whose added metric and mixed terms agree for trace-preserving channels.

The unrestricted complete relative-entropy order $\Phi \succeq_D^c \Psi$ compares

$$D((\Phi \otimes \text{id})(\rho) \| (\Phi \otimes \text{id})(\sigma)) \geq D((\Psi \otimes \text{id})(\rho) \| (\Psi \otimes \text{id})(\sigma))$$

for every finite reference and every state pair, including the usual infinite support values. In particular, equal reference marginals are *not* required. The complete less-noisy order $\Phi \succeq_{\text{ln}}^c \Psi$ compares $I(U : BR)$ for every finite classical ensemble on AR .

Proposition 2.1 (Supplied complete-order equivalence). *For finite-dimensional channels,*

$$\Phi \succeq_{\text{SLD}}^c \Psi \iff \Phi \succeq_D^c \Psi \iff \Phi \succeq_{\text{ln}}^c \Psi.$$

The unrestricted reverse implication from relative entropy to SLD is the saturated-reference filter theorem proved in Section 2 of the supplied companion manuscript [1]. It is not attributed here to the public paper [3], which supplies the forward SLD comparison and the entropic-order framework.

The distinction matters for the negative direction: failure of SLD order must give failure of complete less-noisy order, not merely the failure of a sufficient zero-capacity criterion.

2.2 Gapped cross kernels

Let H_0, H_1 have dimensions h_0, h_1 . Let $A_i > 0$ and $B_i > 0$ be rational effects on the two sectors, with

$$\sum_{i=1}^m A_i = I_{H_0}, \quad \sum_{i=1}^m B_i = I_{H_1}.$$

For $s \in \{E, B\}$, let $O_i^s : H_0 \rightarrow H_1$ be rational rectangular matrices. Outcomes are shared between the two numerator families; an inactive numerator and every common filler are zero. For positive definite R_0 on $H_0\mathbf{R}$, positive definite R_1 on $H_1\mathbf{R}$, and $Z : H_1\mathbf{R} \rightarrow H_0\mathbf{R}$, define

$$\begin{aligned} a_i &= \text{Tr}_{H_0}[(A_i \otimes I)R_0], & b_i &= \text{Tr}_{H_1}[(B_i \otimes I)R_1], \\ o_i^s(Z) &= \sum_{a,b} (O_i^s)_{ba} Z_{ab}, & P_i &= \begin{pmatrix} a_i & 0 \\ 0 & b_i \end{pmatrix}, \\ w_i^s(Z) &= \begin{pmatrix} 0 & o_i^s(Z) \\ o_i^s(Z)^* & 0 \end{pmatrix}, & F_s(R_0, R_1; Z) &= \sum_i g_{P_i}(w_i^s(Z)). \end{aligned} \quad (4)$$

In these conventions,

$$g_{\text{diag}(a,b)} \begin{pmatrix} 0 & o \\ o^* & 0 \end{pmatrix} = 4 \langle o, \mathcal{J}_{a,b}^{-1}(o) \rangle_{\text{HS}}. \quad (5)$$

Proposition 2.2 (Gapped source premise). *The companion game-to-inverse-Jordan reduction effectively produces data of the above kind and a positive rational γ , a finite rational $B > 0$, and a common quadratic seminorm V such that:*

1. *In the nonhalting case, for every finite reference, every positive definite R_0, R_1 , and every complex Z ,*

$$F_E - F_B \geq \gamma V, \quad F_E + F_B \leq BV. \quad (6)$$

2. *In the halting case, some finite faithful R_0, R_1 and some Z satisfy $F_E - F_B < 0$.*

The seminorm ignores precisely the physical cross coordinates invisible to all the numerators. The output data and constants do not depend on a halting witness dimension.

2.3 Exact passage to independent wrapper sectors

For clarity, the data used from the companion are those *before* its final scalar-output POVM completion. Its original physical space is $H = H_{\text{old},0} \oplus H_{\text{old},1}$. It supplies strictly positive block-diagonal effects D_i summing to I_H , and Hermitian original-cross numerators O_i^E, O_i^F . Both quadratures and all common zero-numerator fillers are retained. Make two *new* input sectors, each a copy of H , and put

$$H_0 = H_1 = H, \quad A_i = B_i = D_i, \quad O_i^B = O_i^F.$$

The old and new sector splittings have different roles.

Let C_α be a real Hilbert–Schmidt orthonormal Hermitian basis of the original numerator span. Project the new complex cross Z onto its complexification, writing $Z = \sum_\alpha C_\alpha \otimes Z_\alpha$; its orthogonal complement is invisible. If $W_j = \text{Tr}_H R_j$, choose

$$V(R_0, R_1; Z) = 4 \sum_\alpha \langle Z_\alpha, \mathcal{J}_{W_0, W_1}^{-1}(Z_\alpha) \rangle_{\text{HS}}. \quad (7)$$

Now enlarge the original reference by a two-summand direct sum. Use its input baseline $\text{diag}(R_0, R_1)$ and tangent

$$\begin{pmatrix} 0 & Z \\ Z^* & 0 \end{pmatrix}.$$

Pinching each R_j in the *old* physical split changes neither its D_i outputs nor W_j . The original coefficient tangent is $\begin{pmatrix} 0 & Z_\alpha \\ Z_\alpha^* & 0 \end{pmatrix}$, so its source norm is exactly (7); by (5), its source metric difference is exactly $F_E - F_B$. Thus the original gap is inherited without an extra convention factor. In the companion notation, a tensor-kernel gap η and baseline scaling α_0 give

$$\gamma = \gamma_{\text{cross}} = \frac{2\eta}{\alpha_0}. \quad (8)$$

The quadrature pair is essential to this complexification: it eliminates pseudo-Gram terms and gives the intended sesquilinear form.

An explicit upper frame bound is available as well. Choose rational $d_i > 0$ with $D_i \geq d_i I$. Then $a_i \geq d_i W_0$, $b_i \geq d_i W_1$; inverse Sylvester ordering and coefficient Cauchy–Schwarz give the second inequality of (6), for example with the conservative rational constant

$$B = \max \left\{ \gamma, 2 \sum_{i,s} \frac{\|O_i^s\|_{\text{HS}}^2}{d_i} \right\}.$$

For general A_i, B_i , use $d_i \leq \min\{\lambda_{\min}(A_i), \lambda_{\min}(B_i)\}$. The different noncommutative denominators are retained throughout.

For a negative source witness with unequal reference dimensions n, k , use a new reference of dimension $n + k$. Place the left and right baselines in their respective summands of R_0, R_1 , add faithful positive dummy blocks in the unused summands, and place Z only in the n -by- k corner. The relevant rectangular Sylvester block is unchanged, so the strict negative value transfers exactly. Normalize the full state and tangent together; metric homogeneity preserves its sign.

3 Hub channels and sector-copy padding

Choose a positive rational bound

$$\kappa \geq \max_{i,s} \|B_i^{-1/2} O_i^s A_i^{-1/2}\|_{\text{op}}.$$

Such a bound is computable from rational strictly positive matrices. Choose a positive rational r satisfying

$$r\kappa \leq \frac{1}{2}, \quad r^2 \leq \frac{3\gamma}{16\kappa^2 B}. \quad (9)$$

The all-zero numerator case may be handled separately and is trivial.

For a block input $\rho = [\rho_{jk}]$ on $(H_0 \oplus H_1)\mathbf{R}$, define the normalized hub channels

$$(\mathcal{H}_s \otimes \text{id})(\rho)_i = \begin{pmatrix} \mathcal{A}_i(\rho_{00}) & r o_i^s(\rho_{01}) \\ r o_i^s(\rho_{01})^* & \mathcal{B}_i(\rho_{11}) \end{pmatrix}. \quad (10)$$

Their Gram matrices are

$$G_i^s = \frac{1}{2} \begin{pmatrix} A_i & r(O_i^s)^* \\ r O_i^s & B_i \end{pmatrix} > 0. \quad (11)$$

These ensure complete positivity, and the two effect sums ensure trace preservation. Choose $K_i^s = [F_i, U_i^s]$ with $(K_i^s)^* K_i^s = G_i^s$. The normalized genuine leaf complement of $\mathcal{H}_s$ is

$$\mathcal{L}_s(\rho) = \bigoplus_i 2K_i^s D(K_i^s)^*, \quad D = \text{diag}(\rho_{00}, \rho_{11}). \quad (12)$$

It discards new-sector coherence but retains the residual physical input within each sector. Section 7 realizes these factors rationally in a single balanced isometry.

The sector-copy isometry is

$$V_D |j, a\rangle = |j\rangle_B |j, a\rangle_E.$$

Write $\mathcal{D}$ for its receiver channel and $\mathcal{D}^c$ for its complement. On an input with reference, their outputs are respectively $\text{diag}(\text{Tr}_{H_0} \rho_{00}, \text{Tr}_{H_1} \rho_{11})$ and $\text{diag}(\rho_{00}, \rho_{11})$. Hereafter the untouched reference is suppressed from channel notation.

Lemma 3.1 (Exact padding quotient). *For a faithful ρ and Hermitian tangent X , put*

$$\begin{aligned} \sigma_j &= \text{Tr}_{H_j} \rho_{jj}, & L_j &= \Omega_{\sigma_j}^{-1}(\text{Tr}_{H_j} X_{jj}), \\ Z_j &= X_{jj} - \Omega_{\rho_{jj}}(I_{H_j} \otimes L_j). \end{aligned} \quad (13)$$

Then $\text{Tr}_{H_j} Z_j = 0$, and

$$\Delta_D(\rho, X) := g_{\mathcal{D}^c(\rho)}(\mathcal{D}^c(X)) - g_{\mathcal{D}(\rho)}(\mathcal{D}(X)) = \sum_j g_{\rho_{jj}}(Z_j) \geq 0. \quad (14)$$

Its kernel consists of arbitrary new-sector cross tangents together with the two sectorwise reference-only scores.

Proof. Partial trace sends $\Omega_{\rho_{jj}}(I \otimes L_j)$ to $\Omega_{\sigma_j}(L_j)$, proving the zero marginal. The SLD inner product of Z_j with that removed tangent is $\text{Tr}[Z_j(I \otimes L_j)] = 0$. Its squared norm is $\text{Tr}[\rho_{jj}(I \otimes L_j)^2] = g_{\sigma_j}(\text{Tr}_{H_j} X_{jj})$. Orthogonal decomposition and additivity over sector blocks prove (14). On faithful bases, the residual norms vanish exactly when both residuals vanish. Both channels ignore cross tangents. $\square$

A scalar population pad would miss the two independent reference scores. The full quotient is necessary below.

4 Uniform control of coherent hub baselines

Define

$$\Delta_H(\rho, X) = g_{\mathcal{H}_E(\rho)}(\mathcal{H}_E(X)) - g_{\mathcal{H}_B(\rho)}(\mathcal{H}_B(X)).$$

The source gap applies to the corrected physical cross

$$Z_\times = X_{01} - \frac{(I \otimes L_0)\rho_{01} + \rho_{01}(I \otimes L_1)}{2}. \quad (15)$$

Lemma 4.1 (Coherent hub bound). *Under the positive source hypothesis (6), every finite reference, faithful ρ , and Hermitian X satisfy*

$$\Delta_H(\rho, X) \geq r^2 \left[\frac{\gamma}{4} V(\rho_{00}, \rho_{11}; Z_\times) - C \Delta_D(\rho, X) \right], \quad C = 4\kappa^2 + \frac{8\kappa^2 B}{\gamma}. \quad (16)$$

No constant depends on the smallest eigenvalue of ρ or the reference dimension.

Proof. For any score $\mathcal{L}$, the metric identity is

$$g_\omega(Y) = g_\omega(Y - \Omega_\omega \mathcal{L}) + 2 \text{Tr}(Y \mathcal{L}) - \text{Tr}(\omega \mathcal{L}^2).$$

On each hub subtract $\mathcal{L} = \text{diag}(L_0, L_1)$. The last two terms depend only on the common sector-diagonal output blocks and cancel between E and B . The residual tangent is $d_i + r w_i^s(Z_\times)$, where

$$d_i = \text{diag}(\mathcal{A}_i(Z_0), \mathcal{B}_i(Z_1)).$$

The baseline is $P_i + r Q_i^s$, with P_i formed from ρ_{00}, ρ_{11} and $Q_i^s = w_i^s(\rho_{01})$. Complete positivity of the Gram block with cross coefficient $1/\kappa$ gives

$$-\kappa P_i \leq Q_i^s \leq \kappa P_i.$$

Whiten in Hilbert–Schmidt operator space:

$$u_i = \Omega_{P_i}^{-1/2} d_i, \quad z_i^s = \Omega_{P_i}^{-1/2} w_i^s(Z_\times), \quad T_i^s = \Omega_{P_i}^{-1/2} \Omega_{Q_i^s} \Omega_{P_i}^{-1/2}.$$

Jordan positivity implies $\|T_i^s\| \leq \kappa$. Hub parity decomposes operators into even sector-diagonal and odd sector-cross subspaces. The self-adjoint T_i^s exchanges these subspaces, while u_i is even and z_i^s odd. A block Schur complement gives the exact identity

$$g_{P_i + r Q_i^s}(d_i + r w_i^s) = \|u_i\|^2 + r^2 \langle z_i^s - T_i^s u_i, (I - r^2 (T_i^s)^2)^{-1} (z_i^s - T_i^s u_i) \rangle. \quad (17)$$

The resolvent is restricted to the odd subspace. This is a matrix-valued parity identity; no scalar paired-output formula is assumed.

The common $\|u_i\|^2$ terms cancel. Measurement data processing for (A_i) and (B_i) separately gives

$$U^2 := \sum_i \|u_i\|^2 \leq \sum_j g_{\rho_{jj}}(Z_j) = \Delta_D.$$

Set $a = r^2 \kappa^2 / (1 - r^2 \kappa^2)$. The resolvent correction lies between 0 and aI . From (6), the difference of the leading squares is at least

$$\gamma V - 2\kappa \sqrt{2BV} U - \kappa^2 U^2.$$

Discard the nonnegative E -side resolvent correction. The negative B -side correction is bounded in magnitude by

$$2aBV + 2a\kappa^2 U^2.$$

By (9), $a \leq 1/3$ and $2aB \leq \gamma/2$. Finally,

$$2\kappa \sqrt{2BV} U \leq \frac{\gamma}{4} V + \frac{8\kappa^2 B}{\gamma} U^2.$$

Thus the residual difference is at least $\gamma V/4 - [(1+2a)\kappa^2 + 8\kappa^2 B/\gamma] U^2$. Since $1+2a \leq 5/3 < 4$ and $U^2 \leq \Delta_D$, (16) follows. Coherence in ρ_{01} is controlled entirely through T_i^s ; it has not been removed by assumption. $\square$

5 Exact leaf duality for full physical scores

The leaf contribution must be controlled for all sector-diagonal tangents, not just reference-only scores. Let

$$D = \text{diag}(\rho_{00}, \rho_{11}), \quad X_d = \text{diag}(X_{00}, X_{11}), \quad S = \text{diag}(S_0, S_1), \quad S_j = \Omega_{\rho_{jj}}^{-1}(X_{jj}).$$

Lemma 5.1 (Full-score leaf formula). *For a leaf map K with $G = K^*K > 0$, put*

$$C_G = \sqrt{D} G \sqrt{D}, \quad B_G = \frac{i}{2} \sqrt{D} [G, S] \sqrt{D}.$$

Then

$$g_{KDK^*}(KX_dK^*) = \text{Tr}(DSGS) - g_{C_G}(B_G). \quad (18)$$

The formula holds for nonnormalized positive output blocks.

Proof. Use purification $W = K\sqrt{D}$ with derivative $\dot{W} = KS\sqrt{D}/2$. Its state derivative is KX_dK^* and $4\|\dot{W}\|_{\text{HS}}^2 = \text{Tr}(DSGS)$. A horizontal purification is obtained by minimizing $4\|\dot{W} - WA\|_{\text{HS}}^2$ over anti-Hermitian gauges A . The normal equation is

$$C_G A + A C_G = W^* \dot{W} - \dot{W}^* W = \frac{1}{2} \sqrt{D} [G, S] \sqrt{D}.$$

If $J = 2iA$, it becomes $\Omega_{C_G}(J) = B_G$. Orthogonality of horizontal and vertical derivatives subtracts $4\text{Tr}(A^* C_G A) = \text{Tr}(C_G J^2) = g_{C_G}(B_G)$. The minimal purification metric is the SLD metric, proving the identity. $\square$

Because D, S are sector diagonal and the diagonal blocks of G_i^E, G_i^B agree, the first term in (18) cancels between the two families. It need not vanish, nor do the diagonal commutators with S_j need to vanish.

5.1 One common lifted input and its transpose identity

Let F_j be a copy of $H_j \otimes \mathbb{R}$ and $F = F_0 \oplus F_1$. Use column-vectorized purifications $|\sqrt{\rho_{jj}}\rangle\rangle$ and define the actual unitary path

$$\begin{aligned} |\Psi(u)\rangle\rangle &= \sum_{j=0}^1 |j\rangle e^{iuS_j/2} |\sqrt{\rho_{jj}}\rangle\rangle, \\ \Gamma(u) &= \text{Tr}_{\mathbb{R}} |\Psi(u)\rangle\rangle \langle\Psi(u)|, \quad \Gamma = \Gamma(0), \quad \xi = \Gamma'(0). \end{aligned} \quad (19)$$

The sector notation identifies the physical input summand. The unitary acts on $H_j \otimes \mathbb{R}$, and F_j is untouched. Thus Γ is a normalized positive input on $(H_0 \oplus H_1) \otimes F$, and ξ is a traceless Hermitian tangent to a positive state path.

Lemma 5.2 (Common dual hub realization). *For G_i^s as in (11), the same pair (Γ, ξ) satisfies, on $(\text{hub}_0 \otimes F_0) \oplus (\text{hub}_1 \otimes F_1)$,*

$$(\mathcal{H}_s \otimes \text{id}_F)(\Gamma)_i = 2(C_{G_i^s})^T, \quad (\mathcal{H}_s \otimes \text{id}_F)(\xi)_i = 2(B_{G_i^s})^T. \quad (20)$$

The transpose is a whole-output transpose, not a partial transpose.

Proof. Let $U(u) = \text{diag}(e^{iuS_0/2}, e^{iuS_1/2})$. Contraction of the original reference and physical indices yields

$$(\mathcal{H}_s \otimes \text{id}_F)(\Gamma(u))_i = 2[\sqrt{D} U(u)^* G_i^s U(u) \sqrt{D}]^T.$$

At zero this gives the state identity; differentiating gives the tangent identity. For example, the upper-right state block is

$$r[\sqrt{\rho_{11}}(O_i^s \otimes I)\sqrt{\rho_{00}}]^T,$$

and its derivative is

$$\frac{ir}{2} [\sqrt{\rho_{11}}((O_i^s \otimes I)S_0 - S_1(O_i^s \otimes I))\sqrt{\rho_{00}}]^T.$$

These are the transposes of the lower-left blocks of $2C_{G_i^s}$ and $2B_{G_i^s}$ respectively. The diagonal blocks give the corresponding $A_i S_0 - S_0 A_i$ and $B_i S_1 - S_1 B_i$ commutators. This fixes the rectangular adjoint orientation explicitly. $\square$

Corollary 5.3 (Exact leaf-to-hub reversal). *For the normalized leaf channels of (12),*

$$g_{\mathcal{L}_B(D)}(\mathcal{L}_B(X_d)) - g_{\mathcal{L}_E(D)}(\mathcal{L}_E(X_d)) = \Delta_H(\Gamma, \xi). \quad (21)$$

Proof. The normalized leaf map is $\sqrt{2}K_i^s$, so its metric is twice the metric in (18). The common $\text{Tr}(DSG_i^s S)$ terms cancel. Whole transposition preserves SLD, and $g_{2C^T}(2B^T) = 2g_C(B)$. These two normalization factors cancel exactly in the reversal, with the stated B -minus- E leaf sign. $\square$

5.2 The lifted padding seminorm

Lemma 5.4 (Dual padding bound). *For the common lift of (19),*

$$\Delta_D(\Gamma, \xi) \leq \Delta_D(\rho, X). \quad (22)$$

In fact equality holds.

Proof. Write $S_j = I \otimes L_j + J_j$ with L_j from (13). Then

$$Z_j = \Omega_{\rho_{jj}}(J_j), \quad \sum_j \text{Tr}(\rho_{jj} J_j^2) = \Delta_D(\rho, X).$$

The generator $I \otimes L_j$ acts on the original reference, which is traced out in Γ , so it contributes zero to each diagonal lifted tangent. The remaining diagonal tangent is the reduced pure-state tangent generated by $J_j/2$. The pure-state SLD metric and data processing imply

$$g_{\Gamma_{jj}}(\xi_{jj}) \leq \text{Tr}(\rho_{jj} J_j^2).$$

Subtracting the nonnegative reference-marginal metric in the lifted padding quotient proves (22).

To see equality directly, the untouched F_j marginal of the lifted diagonal sector is constant:

$$\text{Tr}_{H_j} \Gamma_{jj}(u) = [\sqrt{\rho_{jj}} U_j(u)^* U_j(u) \sqrt{\rho_{jj}}]^T = \rho_{jj}^T.$$

Thus its marginal tangent is zero. Purification minimization over Hermitian gauges K on the traced original reference gives

$$g_{\Gamma_{jj}}(\xi_{jj}) = \min_K \text{Tr}[\rho_{jj}(S_j - I \otimes K)^2].$$

The normal equation is $\Omega_{\sigma_j}(K) = \text{Tr}_{H_j} X_{jj}$, whose solution is L_j . The minimum is exactly $g_{\rho_{jj}}(Z_j)$. Summing proves equality. $\square$

Remark 5.5 (Boundary continuation). The lifted Γ need not be faithful. Its unitary-derived tangent can have support-to-kernel blocks, but has zero kernel-to-kernel block and finite SLD information. Apply faithful regularization

$$\Gamma_\varepsilon = \frac{\Gamma + \varepsilon I}{\text{Tr}(\Gamma + \varepsilon I)}, \quad \xi_\varepsilon = \frac{\xi}{\text{Tr}(\Gamma + \varepsilon I)}.$$

Equation (3) gives convergence of the finite metrics used here. In particular the hub identity (20) has positive $C_{G_i^s}$ on each displayed supported subspace. After dropping the nonnegative V term, the uniform inequality in Lemma 4.1 therefore extends to (Γ, ξ) , without introducing a conditioning or reference-size constant.

6 The genuine balanced completion

Use the balanced Stinespring routing whose receiver and actual complement are

$$C_\star = \frac{1}{2}\mathcal{H}_B \oplus \frac{1}{2}\mathcal{L}_E, \quad C_\star^c = \frac{1}{2}\mathcal{L}_B \oplus \frac{1}{2}\mathcal{H}_E. \quad (23)$$

The pair is realized by one isometry in Section 7; these are not freely chosen independent output maps. Define $\Delta_\star$ to be its environment-minus-receiver SLD gap.

Proposition 6.1 (Uniform absorption). *In the positive source case,*

$$\Delta_\star(\rho, X) \geq -Cr^2\Delta_D(\rho, X) \quad (24)$$

for every finite reference, faithful ρ , and Hermitian X . Hence, for any rational $0 < \lambda \leq 1/(1 + Cr^2)$, the physically flagged mixture

$$C = (1 - \lambda)\mathcal{D} \oplus \lambda C_\star, \quad C^c = (1 - \lambda)\mathcal{D}^c \oplus \lambda C_\star^c \quad (25)$$

satisfies $C^c \succeq_{\text{SLD}}^c C$.

Proof. The leaves ignore the new-sector cross blocks. Their exact reversal gives

$$\Delta_\star(\rho, X) = \frac{1}{2}\Delta_H(\rho, X) + \frac{1}{2}\Delta_H(\Gamma, \xi). \quad (26)$$

Apply Lemma 4.1 to each term, use Remark 5.5 for the second, drop the two nonnegative source norms, and apply Lemma 5.4. This proves (24). Both mixture outputs retain their branch flags, so additivity of g over direct sums gives

$$\Delta_C = (1 - \lambda)\Delta_D + \lambda\Delta_\star \geq (1 - \lambda - \lambda Cr^2)\Delta_D \geq 0.$$

□

Proposition 6.2 (Exact negative witness). *In the negative source case, the same constructed C satisfies $C^c \not\succeq_{\text{SLD}}^c C$.*

Proof. Use a faithful block-diagonal input with blocks realizing the negative source form and use only its pure cross tangent $X = \begin{pmatrix} 0 & Z \\ Z^\star & 0 \end{pmatrix}$. The tangent is traceless. Padding and all leaf tangents vanish; the coherent hub operators T_i^s vanish as well. Therefore the exact mixture gap is

$$\Delta_C(\rho, X) = \frac{\lambda r^2}{2}(F_E - F_B) < 0. \quad (27)$$

The parameters were fixed from the source gap data, before any witness was known. There is no halting-dependent small-coupling limit. □

7 Rational Stinespring data and dimensions

Lemma 7.1 (Rational Gram factors). *Every positive rational complex Hermitian q -by- q matrix has a computable rectangular rational complex Gram factor using at most $4q$ rows.*

Proof. Rational $LDL^\star$ elimination gives positive rational diagonal pivots. Each positive rational a/b is a sum of four rational squares: represent the positive integer ab as a sum of four integer squares and divide their roots by b . A finite exhaustive search finds such a representation. Factor every pivot in this way and multiply by the rational triangular factor. No efficiency bound is needed. □

Put $\tau = r/2$. Choose rational F_i with $F_i^* F_i = A_i/2$, using at most $4h_0$ rows. Define in its old leaf coordinates

$$U_i^{s,\text{old}} = 2\tau F_i A_i^{-1} (O_i^s)^*.$$

Then

$$(U_i^{s,\text{old}})^* U_i^{s,\text{old}} = 2\tau^2 O_i^s A_i^{-1} (O_i^s)^*.$$

Append a rational Gram factor of

$$B_i/2 - 2\tau^2 O_i^s A_i^{-1} (O_i^s)^* > 0,$$

using at most $4h_1$ additional rows, and extend F_i by zeros there. After zero padding to a common leaf size $\ell = 4(h_0 + h_1)$, the factors satisfy

$$(U_i^s)^* U_i^s = B_i/2, \quad F_i^* U_i^s = \tau (O_i^s)^*. \quad (28)$$

These are exactly the Grams (11).

Both star output spaces are

$$\bigoplus_{i=1}^m (\mathbb{C}|\text{old}_i\rangle \oplus \mathbb{C}|\text{new}_i\rangle \oplus \text{leaf}_i), \quad n_\star = m[2 + 4(h_0 + h_1)].$$

Define the rational isometry on the two input sectors by

$$\begin{aligned} V_0|a\rangle &= \sum_i (|\text{old}_i\rangle_B \otimes F_i|a\rangle_E + F_i|a\rangle_B \otimes |\text{old}_i\rangle_E), \\ V_1|b\rangle &= \sum_i (|\text{new}_i\rangle_B \otimes U_i^B|b\rangle_E + U_i^E|b\rangle_B \otimes |\text{new}_i\rangle_E). \end{aligned} \quad (29)$$

The ordered tensor supports are disjoint. Equation (28) and the effect sums give

$$V_0^* V_0 = I_{H_0}, \quad V_1^* V_1 = I_{H_1}, \quad V_0^* V_1 = 0.$$

Taking the two partial traces yields precisely (23): receiver hub numerators are O_i^B and environment hub numerators are O_i^E . The actual leaf blocks are the unnormalized $K_i^s D K_i^{s*}$, explaining the half weights in (23).

Choose λ with rational amplitudes by taking an integer $n > 1$ and setting

$$\sqrt{\lambda} = \frac{2n}{n^2 + 1}, \quad \sqrt{1 - \lambda} = \frac{n^2 - 1}{n^2 + 1}. \quad (30)$$

The bound $B \geq \gamma$ and (9) give $Cr^2 \leq 3\gamma/(4B) + 3/2 \leq 9/4$. Thus $n = 4$ suffices uniformly: $\lambda = 64/289$, $\sqrt{\lambda} = 8/17$, and $\sqrt{1 - \lambda} = 15/17$. Combining the copy and star isometries in disjoint output summands gives the genuine flagged pair (25), with rational coefficients. Its input, receiver, and environment dimensions are respectively

$$h_0 + h_1, \quad 2 + n_\star, \quad h_0 + h_1 + n_\star. \quad (31)$$

Thus its Kraus and Choi data are exactly rational complex, not approximations to algebraic data. The qubit half-erasure factor in (1) has rational Choi entries, so $T_{M,x}$ does also.

8 From complete environmental order to capacity

Define the finite-dimensional half-erasure channel by

$$\mathcal{E}_{d,1/2}(\rho) = \frac{1}{2}\rho \oplus \frac{1}{2} \text{Tr}(\rho) |\perp\rangle\langle\perp|.$$

Its genuine complement is isometrically equivalent to the same channel. The next characterization concerns *every* finite helper dimension, before fixing a qubit helper.

Proposition 8.1 (Finite-helper characterization). *For every finite channel C ,*

$$C^c \succeq_D^c C \iff P(C \otimes \mathcal{E}_{d,1/2}) = 0 \text{ for every finite } d. \quad (32)$$

This is Proposition 4.3, equation (4.7), of Hirche [3]; it is not a fixed-helper or quantum-capacity assertion.

Proof. Complete less-noisy comparisons tensorize, treating all other systems as references. Together with the self-complementary helper this proves the forward direction at every blocklength. For the converse, a violating finite ensemble $\{p_x, \rho_x^{\text{AR}}\}_{x \in \mathbb{Z}_k}$ has $\delta = I(X : \text{BR}) - I(X : \text{ER}) > 0$. For uniform $U \in \mathbb{Z}_k$ prepare

$$\sigma_u^{\text{ARK}} = \sum_x p_x \rho_x^{\text{AR}} \otimes |x - u\rangle\langle x - u|_K.$$

The AR marginal is independent of u . Sending RK through half erasure, with $d = k \dim R$, gives

$$I(U : B) = I(U : E) = 0, \quad I(U : \text{BRK}) - I(U : \text{ERK}) = \delta,$$

and private information $\delta/2 > 0$. The order reference is physically transmitted. $\square$

Lemma 8.2 (Private-to-quantum activation identity). *Suppose a finite ensemble $\{p_x, \rho_x^A\}$ has private information $\delta > 0$ through $L : A \rightarrow B$, with genuine environment E . Then $L \otimes \mathcal{E}_{e,1/2}$ has a positive single-use coherent-information witness for some finite e .*

Proof. Choose purifications $|\psi_x\rangle^{AF_2}$ and let $F = F_1 F_2$, with orthogonal x -flags in F_1 . Purify the flagged mixture as

$$|\Phi\rangle^{\text{RAF}} = \sum_x \sqrt{p_x} |x\rangle_R |x\rangle_{F_1} |\psi_x\rangle^{AF_2}.$$

Send A through L and F through half erasure. If F survives, $S(BF) = H(p) + \sum_x p_x S(E_x)$ and Eve's entropy is $S(E_{\text{av}})$. If erased, Bob's entropy is $S(B_{\text{av}})$ and $S(EF) = H(p) + \sum_x p_x S(B_x)$. The flag entropies cancel, yielding the exact Smith–Yard identity [4]

$$I_c(L \otimes \mathcal{E}_{\dim F, 1/2}, \text{Tr}_R |\Phi\rangle\langle\Phi|) = \frac{1}{2}[I(X : B) - I(X : E)] = \delta/2 > 0. \quad (33)$$

One may take $\dim F \leq k \dim A$ for k ensemble letters. $\square$

Thus failure of complete environmental order supplies positive coherent information for C together with *two* finite half-erasure helpers. Their dimensions depend on a finite witness. The following code lemma removes this dependence from the channel instance.

9 Zero-rate compilation into qubit half erasure

Lemma 9.1 (Fixed logical helper simulation). *Fix a logical dimension d . There are finite isometries $W_m : \mathbb{C}^d \rightarrow (\mathbb{C}^2)^{\otimes m}$ and receiver-only decoders Dec_m such that*

$$T_m = \text{Dec}_m \circ (\mathcal{E}_{2,1/2})^{\otimes m} \circ W_m, \quad \|T_m - \mathcal{E}_{d,1/2}\|_{\diamond} \rightarrow 0. \quad (34)$$

Every rejected erasure mask is retained as an explicit failure outcome. The rate $\log_2 d/m$ tends to zero.

Proof. Haar second moment. Let a Haar-random isometry $W : \mathbb{C}^d \rightarrow B \otimes E$ have $b = \dim B$, $e = \dim E$, and $D = be \geq d$, with $D > 1$. Encode half of a normalized maximally entangled logical state, and denote the reduced state on the logical reference and E by ω_{RE} . Twirling the two total-output copies gives identity and swap coefficients on the two reference copies

$$A_{RR'} = \frac{I_{RR'} - F_{RR'}/D}{d^2(D^2 - 1)}, \quad B_{RR'} = \frac{F_{RR'} - I_{RR'}/D}{d^2(D^2 - 1)}.$$

Indeed, the two twirl coefficients are fixed by its total trace and its trace against the total-output swap; for a normalized maximally entangled input those reference operators are I/d^2 and F/d^2 , respectively. Tracing against the relevant swaps yields

$$\begin{aligned} \mathbb{E} \text{Tr} \omega_{RE}^2 &= \frac{e(b^2 - 1)/d + b(e^2 - 1)}{b^2 e^2 - 1}, \\ \mathbb{E} \text{Tr} \omega_E^2 &= \frac{e(b^2 - 1) + b(e^2 - 1)/d}{b^2 e^2 - 1}. \end{aligned} \quad (35)$$

Put $\pi_R = I_R/d$ and $\Delta_{RE} = \omega_{RE} - \pi_R \otimes \omega_E$. The cross term in its squared norm is $\text{Tr} \omega_E^2/d$, so

$$\mathbb{E} \text{Tr} \Delta_{RE}^2 = \frac{b(e^2 - 1)(1 - d^{-2})}{b^2 e^2 - 1}. \quad (36)$$

Its rank is at most de . Cauchy–Schwarz, Jensen, and $b^2(e^2 - 1) \leq b^2 e^2 - 1$ therefore give

$$\mathbb{E} \|\omega_{RE} - \pi_R \otimes \omega_E\|_1 \leq \sqrt{(d - 1/d)e/b}. \quad (37)$$

One code for all accepted masks. Encode into m qubits. Every erasure mask has input-independent probability 2^{-m} . A mask with s survivors has $b = 2^s$, $e = 2^{m-s}$. Let

$$g(m) = \lceil m^{1/4} \rceil, \quad \text{accept a mask exactly when } s \geq m/2 + g(m).$$

On each accepted mask, (37) is at most $\sqrt{d} 2^{-g(m)}$. Average over masks and then over W . There is consequently one isometry W_m whose probability-weighted accepted-mask decoupling error is at most this same bound. This does not choose a different encoding for each mask.

For each accepted mask, Uhlmann recovery gives a trace-preserving decoder with entanglement infidelity at most its decoupling trace error. The corresponding normalized Choi trace distance is at most twice the square root of that error. Jensen’s inequality over masks bounds the weighted Choi error by $2d^{1/4} 2^{-g(m)/2}$. The Choi-to-diamond estimate multiplies this by d .

Rejected masks output the fixed orthogonal flag $|\perp\rangle$. Accepted decoders output the logical d -dimensional system. If

$$p_m = \Pr\{\text{Binomial}(m, 1/2) \geq m/2 + g(m)\},$$

define $\mathcal{E}_d^{(p)}(\rho) = p\rho \oplus (1-p) \text{Tr}(\rho) |\perp\rangle\langle\perp|$, with p the *success* probability. The decoded channel satisfies

$$\|T_m - \mathcal{E}_d^{(p_m)}\|_\diamond \leq 2d^{5/4}2^{-g(m)/2}. \quad (38)$$

The mask probability is input-independent, so this is an honest channel estimate, not a conditioned-success comparison. Since $g(m) \rightarrow \infty$ and $g(m)/\sqrt{m} \rightarrow 0$, the binomial central limit theorem gives $p_m \rightarrow 1/2$. Finally,

$$\|T_m - \mathcal{E}_{d,1/2}\|_\diamond \leq 2d^{5/4}2^{-g(m)/2} + 2|p_m - 1/2| \rightarrow 0. \quad (39)$$

The logical d is fixed before increasing m , proving the zero-rate assertion. $\square$

No computational bound on the resulting code length is used in the channel reduction. Existence of a finite sufficiently accurate code is enough for positivity. If a concrete code realization is required, isometries and finite-dimensional recovery maps admit dense computable parametrizations, and strict fidelity targets can be certified in finite dimension. This observation is not an efficiency claim.

10 The fixed-qubit capacity equivalence

Proposition 10.1 (Fixed qubit helper). *For every finite channel C ,*

$$C^c \succeq_D^c C \iff P(C \otimes \mathcal{E}_{2,1/2}) = 0 \iff Q(C \otimes \mathcal{E}_{2,1/2}) = 0. \quad (40)$$

Proof. Complete environmental order gives $P = 0$ by Proposition 8.1 with $d = 2$, and $Q \leq P$ gives $Q = 0$.

If complete order fails, Proposition 8.1 gives a finite one-use private witness for $C \otimes \mathcal{E}_{d,1/2}$. Apply Lemma 8.2 to this product channel. It gives a fixed input state with strictly positive coherent information for

$$C \otimes \mathcal{E}_{d,1/2} \otimes \mathcal{E}_{e,1/2}$$

for some finite d, e . Choose a purification R of this fixed input. Compile the two logical helpers independently by Lemma 9.1, with finite blocklengths m_1, m_2 . Use $m = m_1 + m_2$ copies of $C \otimes \mathcal{E}_{2,1/2}$. One C input carries the witness's C system; every other C input is a fixed pure state. The first and second helper inputs are isometrically encoded into their respective qubit blocks. All receiver helper outputs are locally decoded, and spare receiver C outputs can be discarded.

The decoded receiver channel converges in diamond norm to the ideal finite product channel. On the fixed purification, coherent information is

$$I_c = -S(R|B).$$

Conditional-entropy continuity in the fixed reference dimension shows that it remains strictly positive for sufficiently accurate *finite* helper codes. Receiver postprocessing cannot increase coherent information, so the original, pre-decoder physical block also has positive coherent information. A spare pure C input contributes zero coherent information; no auxiliary resource is supplied for free. The regularized coding theorem [5] gives

$$Q(C \otimes \mathcal{E}_{2,1/2}) \geq \frac{1}{m} I_c((C \otimes \mathcal{E}_{2,1/2})^{\otimes m}, \rho_{\text{code}}) > 0.$$

Then $P > 0$ as well, proving both converses. $\square$

This proof applies continuity *after* private-to-quantum activation. It never identifies the physical lost-qubit subsystem with an ideal helper complement. The identity $I_c = -S(R|B)$ and receiver data processing automatically include any environment created by the decoder. Direct receiver-channel simulation by itself would not justify a private-information comparison that ignored that environment.

11 Halting reduction and recursive enumerability

The source and Propositions 6.1–6.2 effectively construct $C_{M,x}$ with

$$\begin{aligned} M(x) \text{ nonhalting} &\implies C_{M,x}^c \succeq_{\text{SLD}}^c C_{M,x}, \\ M(x) \text{ halting} &\implies C_{M,x}^c \not\succeq_{\text{SLD}}^c C_{M,x}. \end{aligned}$$

Proposition 2.1 transfers this exact dichotomy to unrestricted complete environmental relative-entropy order. Proposition 10.1 then proves the capacity dichotomy of Theorem 1.1 for $T_{M,x}$ of (1).

If $h = \dim H$ in the source wrapper and $n_\star = m(2 + 8h)$, this target has input dimension $4h$ and receiver dimension $3(2 + n_\star)$. One environmental realization has dimension $3(2h + n_\star)$; no minimality is claimed. All channel entries are computable rational complex. The positive code and its finite reference may be extremely large, but they are witnesses for finite blocks of this one ordinary memoryless channel. There is no catalyst, preshared entanglement, feedback, or postselected capacity convention.

Lemma 11.1 (Positive capacities are recursively enumerable). *On valid finite rational Choi descriptions, $\{\Phi : Q(\Phi) > 0\}$ and $\{\Phi : P(\Phi) > 0\}$ belong to RE.*

Proof. From a rational Choi matrix one computes a finite complementary channel, using a rational rectangular Gram factor as in Lemma 7.1 or a computable algebraic Stinespring factor. For a semidefinite Choi matrix, pivoted rational LDL^* omits its zero pivots and gives the same effective factorization. Finite-dimensional entropy is computable: eigenvalues can be approximated with certification, and $-t \log_2 t$ is continuous at zero.

For Q , enumerate every $n \geq 1$, rational complex density matrix on the n inputs, and precision of a certified entropy approximation. Halt whenever a strictly positive lower bound on its coherent information is obtained. The regularized formula implies that $Q > 0$ has some finite-block positive witness. Density of rational states and continuity of finite-dimensional entropy give a nearby rational witness, so this search terminates exactly on positive-capacity channels.

For P , enumerate every n , every finite ensemble size, rational priors, and rational complex input density matrices, and certify positivity of $I(X : B^n) - I(X : E^n)$ in the same way. A positive private capacity has a finite positive-information ensemble witness; density and continuity again ensure eventual detection. No finite alphabet bound is needed because all alphabet sizes are enumerated. The regularized formulas used here are the usual unassisted coding formulas [5]. $\square$

The halting map makes each positive set RE-hard. Lemma 11.1 makes them RE-complete, and their exact-zero complements are therefore coRE-complete. This completes the proof of the classification under the stated companion source theorems.

12 Scope and verification status

The completed reduction is an exact qualitative test. It does not imply noncomputability of arbitrary-precision numerical capacity approximation or of a fixed-gap promise problem. The positive-capacity witness can occur at an unknown, arbitrarily large blocklength; no computable witness bound, practical code construction, or polynomial-time reduction is claimed. The zero-rate helper code transports a fixed finite logical system with vanishing error while using increasingly many helper qubits. It does not transmit quantum information at positive rate through the half-erasure channel alone.

The new completion applies to the supplied *uniformly gapped* source family. It is not a compiler for every arbitrary ungapped POVM pair. All actual reference quantifiers in its soundness proof are unrestricted and finite; no equal-reference-marginal restriction, scalar-denominator replacement, realification, or state condition-number assumption is used.

This is a research proof draft. Its components and assembled reduction have received internal analytic checks; external review remains pending. The classification relies on the explicitly supplied companion source theorem and complete-order equivalence, together with the complementary completion and helper proofs given here. No author attribution, peer-review claim, or historical originality claim is made.

References

- [1] *Undecidability of complete quantum channel information order*. Companion research manuscript supplied with this work, dated 10 October 2026. In particular, Section 2 proves the unrestricted complete SLD–Umegaki equivalence; Sections 3–8 give the game-to-gapped-kernel and numerator realization used here. This is not an external publication.
- [2] Zhengfeng Ji, Anand Natarajan, Thomas Vidick, John Wright, and Henry Yuen. *MIP*=RE*. arXiv:2001.04383. <https://arxiv.org/abs/2001.04383>.
- [3] Christoph Hirche. *Quantum estimation, channel orders, and private capacity*. arXiv:2610.03706v1 (2026), especially Proposition 4.3 and Section 3. <https://arxiv.org/html/2610.03706v1>.
- [4] Graeme Smith and Jon Yard. *Quantum Communication With Zero-Capacity Channels*. Science **321**, 1812–1815 (2008). arXiv:0807.4935. <https://arxiv.org/abs/0807.4935>.
- [5] Igor Devetak. *The private classical capacity and quantum capacity of a quantum channel*. IEEE Transactions on Information Theory **51**, 44–55 (2005). arXiv:quant-ph/0304127. <https://arxiv.org/abs/quant-ph/0304127>.